3D-Stacked Cross-Coupled ZnO/Fe-ZnO TFT Differential Bit-Cell Enabling High-Density Memory and BCAM Hamming-Distance Comparison

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Abstract

We demonstrate a 3D-stacked cross-coupled differential bit-cell integrating ZnO *n*-type FET access devices and ferroelectric ZnO TFT (FeTFT) storage devices for density improvement and associative computing. The access FETs are monolithically stacked above FeTFTs to form a compact differential topology supporting write 1/0 and differential read. Device measurements and compact model fitting validate both ZnO FET and FeTFT characteristics. A binary CAM (BCAM) array is realized by forming a match line (ML) through SL/SLB connections and search inputs on BL/BLB. Measured ML discharge behavior under match/mismatch enables BCAM operation and provides an analog signature that can be mapped to Hamming distance across multiple mismatched bits. We further prototype a vector retrieval flow that maps LLM embedding binarization to BCAM distance compute and demonstrate faster time-to-first-token (TTFT) compared with GPU baseline.

Keywords (optional): Stacked FET, FeTFT, cross-coupled, BCAM, hamming distance

Introduction

High-density memories compatible with 3D integration are increasingly attractive for both storage and in-memory computing [1, 2]. Content-addressable memory (CAM) further enables fast search by comparing stored and input data in-memory. Moreover, Hamming-distance-aware comparison extends CAM toward similarity search and error-tolerant matching [3, 4], which is useful for accelerating knowledge retrieval in large language model (LLM) applications [5].

This work proposes a 3D-stacked cross-coupled differential bit-cell using ZnO *n*-type FET (N1 and N2, in Fig. 1) and ferroelectric ZnO TFT (NF1 and NF2, in Fig. 1), achieving compact storage while enabling binary CAM (BCAM) operation and Hamming-distance esatimation via analog discharge of the match-line (Fig. 1). Fig. 2 outlines the BCAM array concept: BL/BLB carry the search voltages (V_S/V_{SB}), and SL/SLB are connected through an access device to form the match line (ML), where match/mismatch is sensed by ML discharge behavior.

Bit-Cell Structure and Device Fabrication

Fig. 3(a) shows the physical structure of our proposed 3D stacked differential bit-cell, with access FETs on the top layer and FeTFTs on the bottom layer, enabling compact stacking without increasing 2D footprint. Fig. 3(b) and Fig. 3(c) show the TEM image of and the process integration flow for fabricating our proposed stack bit-cell, respectively. Device fabrication starts from the cleaning of SiO_2/Si sample followed by the deposition of 40 nm thick W metal, which forms the gate electrodes of the FeTFTs. Then, the 7/2/7 nm thick HZO/Al_2O_3/HZO interlayer is deposited by ALD. Next, the 7 nm thick ZnO channel layer is deposited by ALD followed by the device isolation. The source/drain electrodes of the FeTFTs are deposited by E-Beam. Then, the Al_2O_3 and SiO_2 insulating layers are deposited by ALD and PECVD, respectively. The ZnO FET fabrication starts on the SiO_2 insulating layer, followed by the deposition of 40 nm thick W metal. Next, the 15 nm thick HfO_2 dielectric layer and 7 nm thick ZnO channel layer are deposited by ALD. Etching of the ZnO channel layer completes isolation. Then, etching the second HfO_2 layer and SiO_2, Al_2O_3 and HZO/Al_2O_3/HZO layers are achieved with ICP. Finally, the source/drain electrodes of the ZnO TFTs are formed and the via of the L1 and L2 filled after the deposition and lift-off process of Ni metal.

Fig. 4 shows cross-sectional and compositional verification of our fabricated devices. Fig. 4(a) shows a TEM cross-section including the ZnO TFT layer (L2) and FeTFT layer (L1). Fig. 4(b, c) show EDX elemental mapping, and Fig. 4(d, e) show elemental distributions along the indicated line cut, verifying material integrity and layer definition required for monolithic stacking.

Results and Discussion

A. Basic Cell Operation: Fig. 5 shows the experiment data of each device in a single bit-cell and the fitting results based on our compact models for the access FET and FeTFT. Fig. 6(a) and (b) show the write operation of the proposed bit-cell. Fig. 6(c) shows the differential read operation achieved by comparing the current difference on BL and BLB. The transient waveform based on the measured data is shown in Fig. 6(d). Fig. 7 further analyzes the read operation. As Fig. 7(a) shows, WL voltage should be at least 1.02 V to compensate for the mismatch between N1 and N2 in the bit-cell. Higher WL voltage increases the current difference (Fig. 7) for improved sensing.

B. Binary CAM operation: As shown in Fig. 2, BL and BLB are given complementary voltages (V_S and V_{SB}) as search input, Fig. 8(a) and (b) show the match results with different inputs under store 1 and store 0 respectively. Fig. 8(c) and (d) shows the ML signals under match/mismatch conditions for store 1 and store 0, respectively. The ML voltage differences between match and mismatch are 105 mV and 67 mV for store 0 and store 1, respectively, which is caused by the device mismatch between NF1 and NF2. To further study the scalability of the proposed BCAM array, Fig. 9(a) shows the voltage drop on BL/BLB under different array sizes, indicating the array can be scaled to 128 rows and maintain the BL voltage >0.9 V. Fig. 9(b) shows the voltage drop on ML for different array sizes, due to low leakage of N1 and N2. ML voltage remain >0.9 V at 256 columns. Fig. 9(c) and (d) demonstrate voltage drops on ML for varying number of mismatched cells in a row with 8 columns, indicating the proposed BCAM can be used for Hamming distance estimation.

C. BCAM Array for Hamming Distance Estimation: Fig. 10 shows the measurement setup. The TEM image confirms the fabricated array, which is wire-bonded to a chip carrier soldered on a PCB with the required peripherals for testing. Fig. 11(a) and Fig. 11(b) show the Hamming-distance-estimation flow and column-wise search for both match and mismatch conditions, respectively. The Hamming-distance estimation can serve as a retriever for LLMs in retrieval augmented generation (RAG) by comparing the encoded query with previously stored embedding vectors in a knowledge database. As Fig. 12 (a) shows, the proposed BCAM can act as a cache for the knowledge database and Hamming distance is estimated in-memory, thereby reducing data movement and improving time-to-first-token (TTFT). Compared to an NVIDIA H100 GPU baseline, Fig. 12 (b) shows that by tiling the multiple arrays of the proposed BCAM to achieve a larger capacity, up to 38.4% and 28.92% faster TTFT for Mistral-7B and Llama 3-8B, respectively, can be achieved.

Conclusion

A 3D stacked ZnO/Fe-ZnO cross-coupled differential bit-cell is demonstrated along with measured device characterization, compact model calibration, validated write/read operation, and BCAM array demonstrations. The scalable array enables in-memory Hamming distance estimation that can accelerate RAG flows in LLM pipelines.

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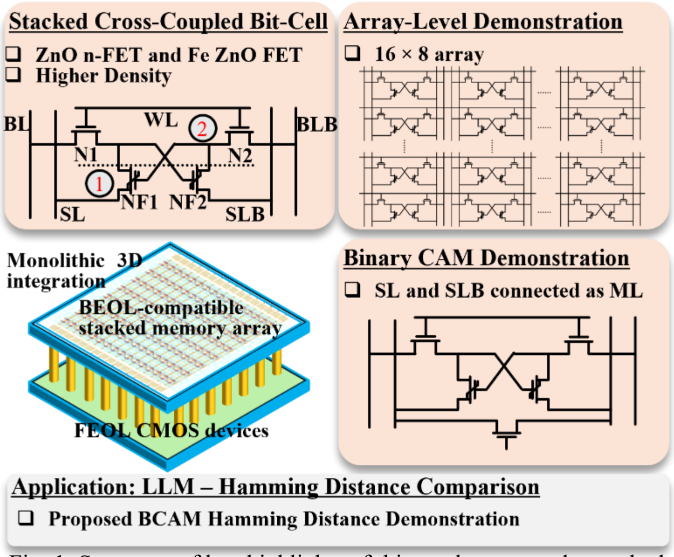

Fig. 1. Summary of key highlights of this work, proposed a stacked cross-coupled bit-cell using ZnO n-FET and Fe ZnO FET for higher density, can be used as BCAM for hamming distance comparison.

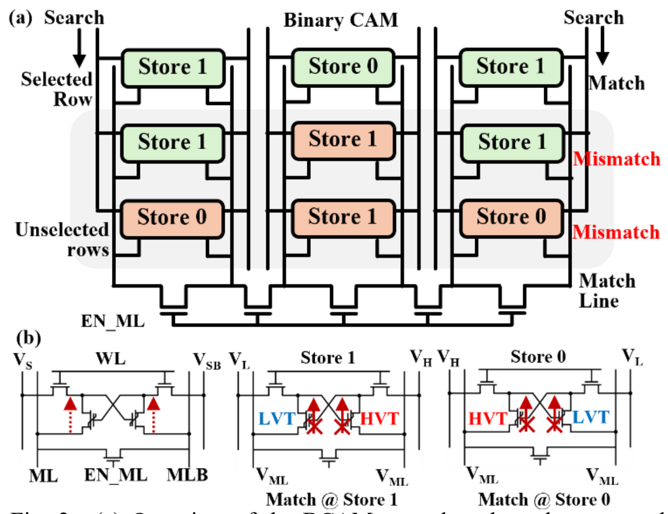

Fig. 2. (a) Overview of the BCAM array based on the proposed differential bit-cell. SL and SLB are connected through an access transistor to form a ML. (b) BCAM cell demonstration and match condition for store 1 and store 0. BL and BLB are used for data search by applying V_S and V_{SB} respectively.

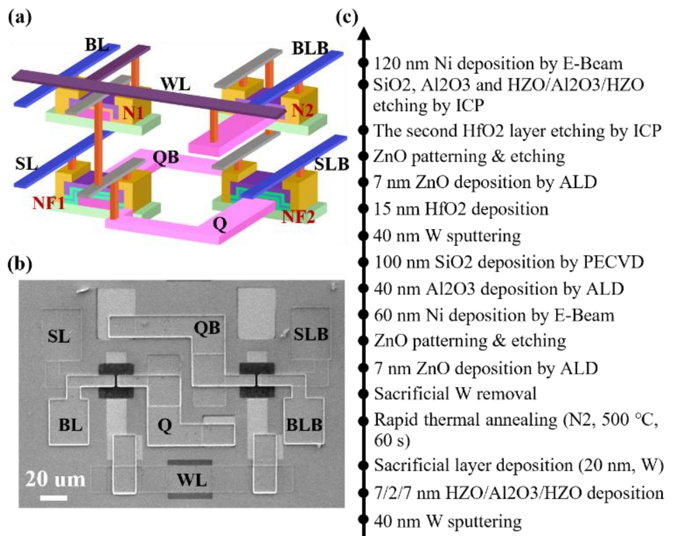

Fig. 3. (a) 3D view of the proposed 3D stacked differential bit-cell with access transistor at the top level and the FeTFT at the bottom level. (b) TEM image of the 3D stacked bit-cell with scale. (c) Fabrication process flow of the 3D stacked bit-cell.

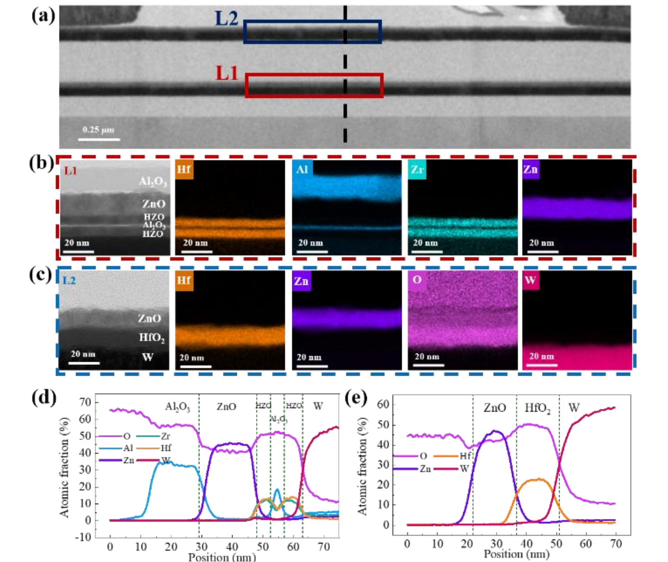

Fig. 4. (a) Cross-sectional TEM image and (b,c) EDX elemental mapping of the ZnO TFT (L2) and FeTFT (L1). (d, e) Elemental distribution along the middle line of (a).

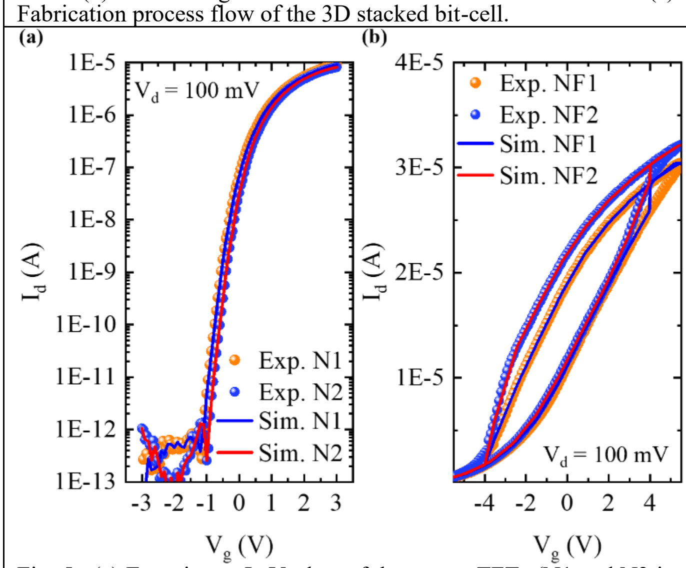

Fig. 5. (a) Experiment I_d-V_g data of the access TFTs (N1 and N2 in the bit-cell) with fitted simulation results of our compact model. (b) Experiment I_d-V_g data of the FeTFTs (NF1 and NF2 in the bit-cell) with fitted simulation results of our compact model.

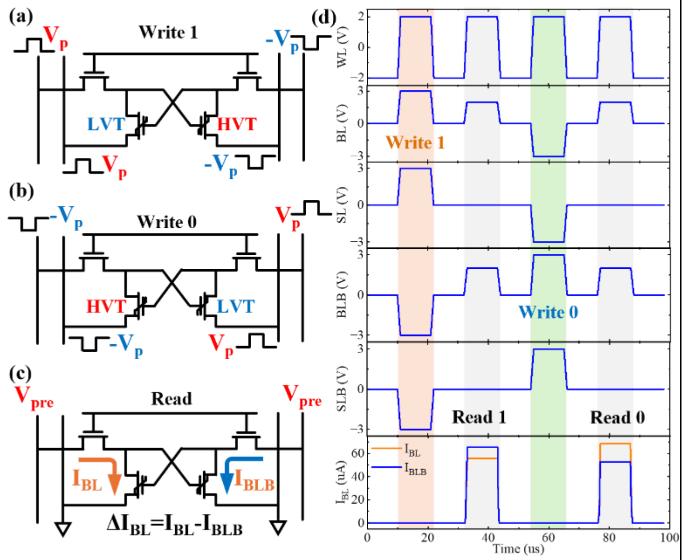

Fig. 6. (a) Write 1 and (b) Write 0 operation of the proposed stacked cross-coupled bit-cell. (c) Differential read operation. (d) Transient waveform of write and read operation.

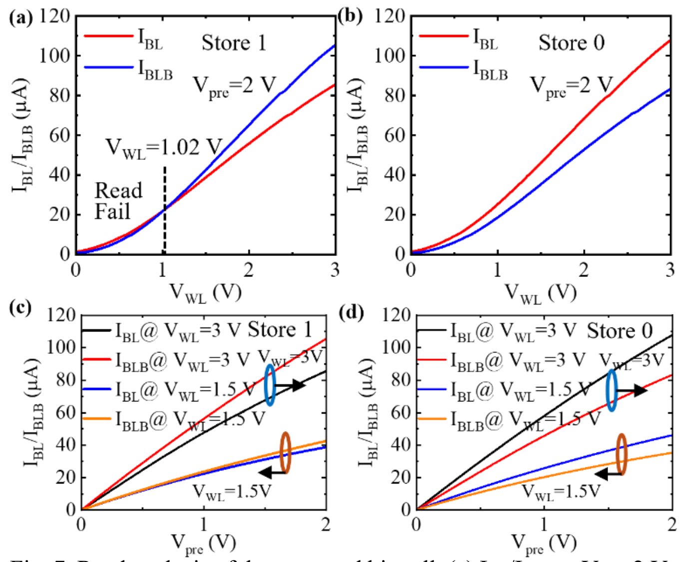

Fig. 7. Read analysis of the proposed bit-cell. (a) I_{BL}/I_{BLB} at V_{pre}=2 V with different WL voltage when store 1. (b) I_{BL}/I_{BLB} at V_{pre}=2 V with different WL voltage when store 0. (c, d) I_{BL}/I_{BLB} at different V_{pre}.

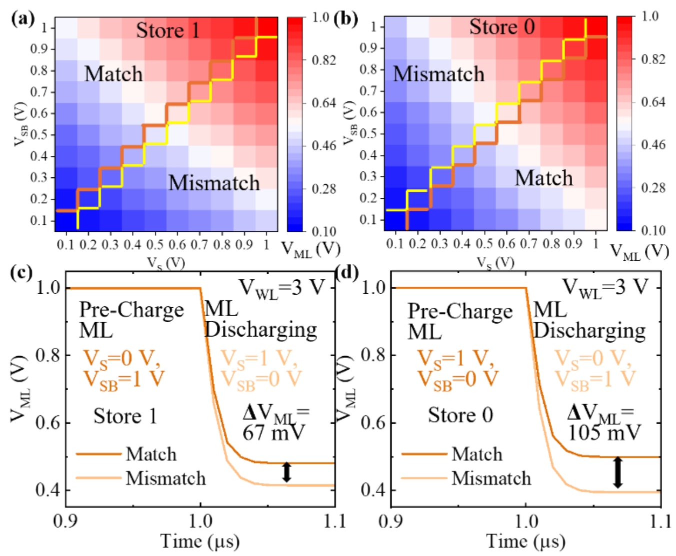

Fig. 8. Binary CAM operation demonstration. (a, b) input voltage analysis on BL/BLB for store 1 and store 0 respectively. (c, d) ML discharging waveform for store 1 and store 0 respectively.

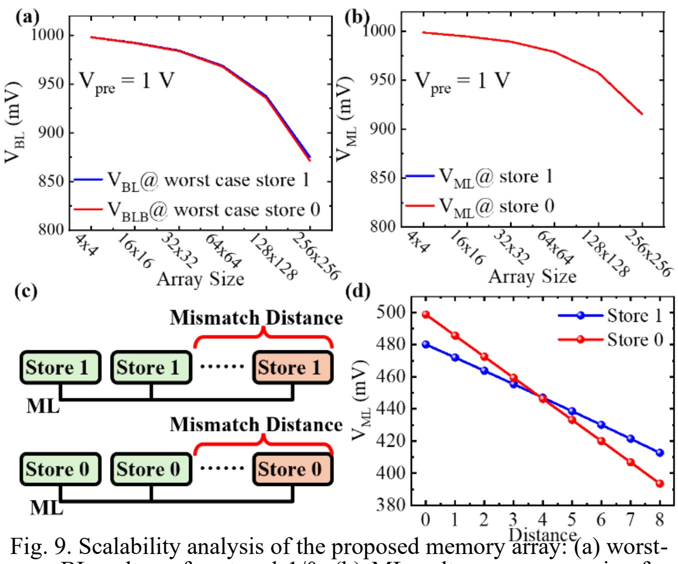

Fig. 9. Scalability analysis of the proposed memory array: (a) worst-case BL voltage for stored 1/0; (b) ML voltage vs. array size for stored 1/0. (c) Distance vs. number of mismatched cells; (d) ML voltage vs. distance for stored 1/0.

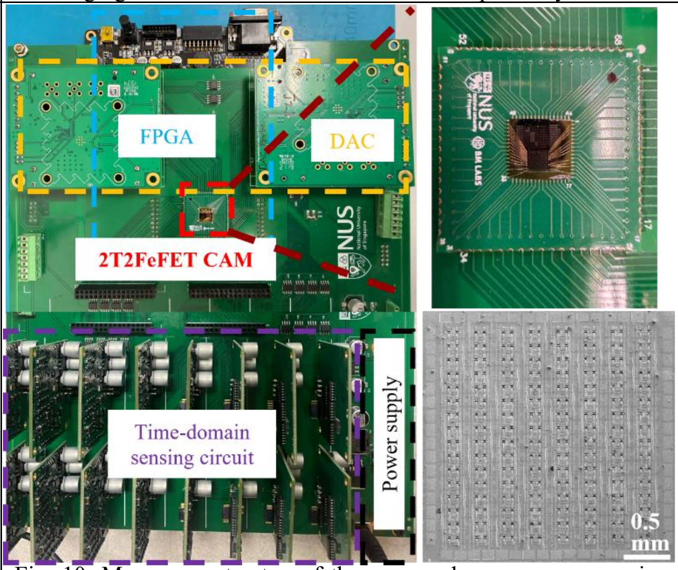

Fig. 10. Measurement setup of the proposed memory array: wire-bonded chip carrier on a PCB with required components, and a TEM image of the array.

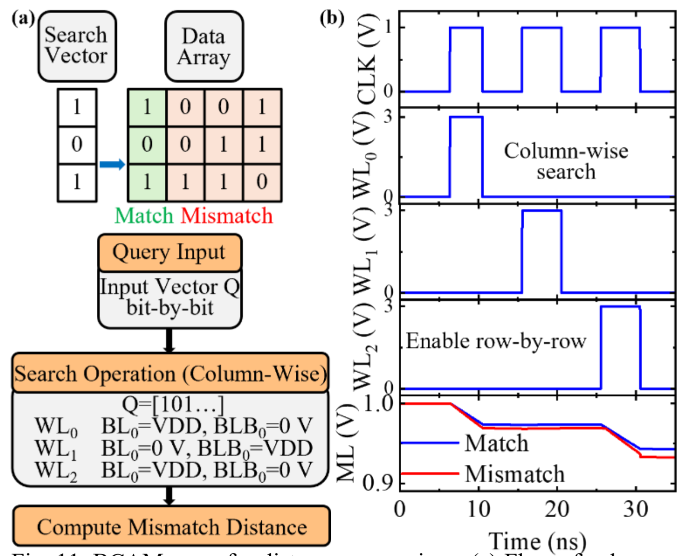

Fig. 11. BCAM array for distance comparison. (a) Flow of column-wise search operation and (b) the transient waveforms during match/mismatch conditions.

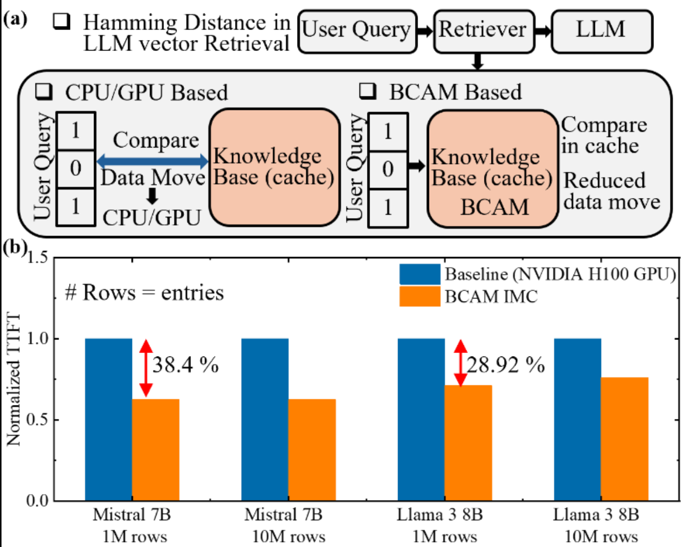

Fig. 12. LLM vector retrieval with BCAM Hamming-distance estimation. (a) BCAM acceleration compared to CPU/GPU. (b) Normalized TTFT vs. model type and BCAM array size.